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Domain and Range

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Domain and Range

You’ve already seen that a function is much like a machine. A number from the domain is substituted for the variable, and a number from the range comes out.

For many functions, the domain and the range will be the set of real numbers, which includes all rational and irrational numbers. Mathematicians use the symbol $\mathbb{R}$ to refer to this set. When the domain and the range equal $\mathbb{R}$, you know that for any real number x , the resulting $f(x)$ will also be a real number.

When you find the domain, you are really watching for a few exceptions to the rules of math that can cause problems.

Remember these restrictions:

You can’t divide by 0.

You can’t find the square root of a negative number.

Example 1

What is the domain of $f(x) = x + 3$?

The domain is $\mathbb{R}$. You can use any real number for x .

Set notation: Domain = $\{x \in \mathbb{R}\}$

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Example 2

What is the domain of $f(x) = \frac{1}{x-2}$?

The domain is **all real numbers except for 2**; $x \neq 2$.
If x equals 2, the denominator of the fraction will be 0.

Domain = $\{x \in \mathbb{R} \mid x \neq 2\}$

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"x is the set of all real numbers such that x can't equal zero."

Restriction:
 $x - 2 \neq 0$
 $x \neq 2$

Example 3

What is the domain of $f(x) = \sqrt{x-2}$?

The domain is **all real numbers greater than or equal to 2**; $x \geq 2$.
If $x < 2$, then the number in the square root symbol will be negative.

Domain = $\{x \in \mathbb{R} \mid x \geq 2\}$

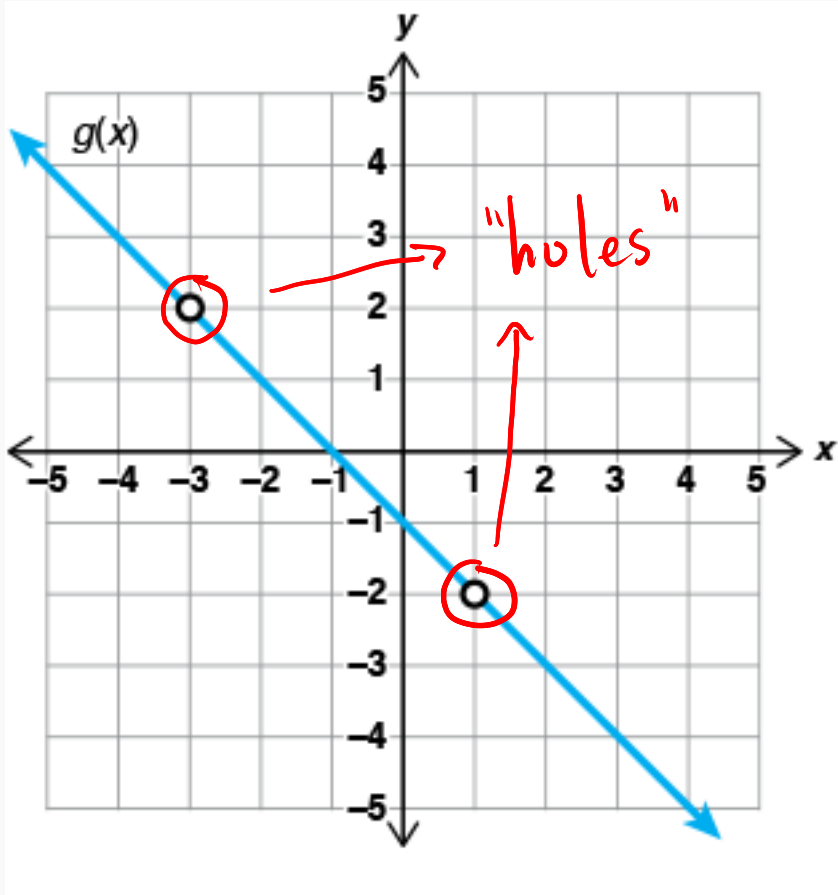
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$x - 2 \geq 0$
 $x \geq 2$

When a number is excluded from the domain, the graph of the function will have an open circle to show that a value is missing.

Example 4

What is the domain and range of $g(x)$?



* For some functions we see a vertical asymptote instead of a hole for missing values in the domain and range.

The domain is **all real numbers except 1 and -3**. If x is equal to 1 or -3, the graph has an open circle to show that there is no corresponding value for y .
The range is **all real numbers except 2 and -2**. There are no values of x that can produce either of these results. The open circles tell you that these values are missing from the range.

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Example 5: $f(x) = \frac{3x-2}{(2x-1)(x+5)}$

Restrictions: $2x-1 \neq 0$
 $2x \neq 1$
 $x \neq \frac{1}{2}$ | $x+5 \neq 0$
 $x \neq -5$

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Category

Lesson 5: Functions

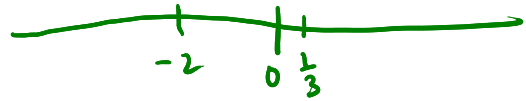
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Practice questions taken related to this topic

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Example 6:



$$g(x) = \frac{5x-2}{x^2(3x-1)(x+2)}$$

Restrictions:

$$x^2 \neq 0$$

$$x \neq 0$$

$$3x-1 \neq 0$$

$$3x \neq 1$$

$$x \neq \frac{1}{3}$$

$$x+2 \neq 0$$

$$x \neq -2$$

$$\text{Domain} = \left\{ x \in \mathbb{R} \mid x \neq 0, -2, \frac{1}{3} \right\}$$

$$\text{Range} = \{ y \in \mathbb{R} \}$$

